

RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering

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MINI PROJECT REPORT

On

Augmented Reality Object Overlay using Feature Matching using Computer Vision and Image Processing

Individual Report — Algorithm : SIFT + BFMatcher + RANSAC Warp

Submitted by

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1. INTRODUCTION

1.1 Background and Motivation

Augmented Reality (AR) is a technology that enhances the real-world environment by overlaying virtual objects onto physical scenes in real time. AR applications are widely used in education, gaming, medical visualization, navigation, and industrial maintenance. Feature-based AR is one of the most reliable approaches, where distinctive image features are detected and matched from images or video frames. In this project, SIFT features are used because of their robustness, scale invariance, and strong support in OpenCV.

1.2 Problem Statement

Traditional AR systems require accurate object localization and alignment to correctly overlay virtual content onto real-world objects. The challenge is to detect and match image features under different positions, scales, rotations, and lighting conditions while maintaining reliable homography estimation and stable virtual object overlay in real time.

1.3 Objectives

The primary objectives of this mini-project are as follows:

1. Generate a synthetic dataset containing reference object images..
2. Detect and extract SIFT feature keypoints using OpenCV.
3. Match image features using BFMatcher.
4. Estimate homography using RANSAC Warp for object alignment.
5. Overlay virtual image content onto the detected object and evaluate matching accuracy.

1.2 Scope of the Report

This report focuses only on the implementation of Augmented Reality Object Overlay using SIFT Feature Detection, BFMatcher Feature Matching, and RANSAC Warp Homography Estimation.

The scope is limited to:

- Offline processing of static RGB images containing reference objects
- Synthetic dataset generation using OpenCV and NumPy
- Use of traditional computer vision and image processing techniques only (no neural networks or deep learning)
- Feature detection and descriptor extraction using SIFT algorithm
- Feature matching using OpenCV BFMatcher module
- Homography estimation using RANSAC Warp transformation
- Virtual image overlay using perspective transformation and object alignment
- Detailed documentation of preprocessing, matching stages, results, and visual outputs.

The following are out of scope:

- Real-time video or webcam-based object tracking
- Deep learning-based object detection or matching methods (YOLO, CNN, etc.)
- 3D object rendering and animation
- Mobile or embedded system deployment
- Markerless augmented reality techniques

This report serves as the individual documentation for the proposed augmented reality pipeline and provides a foundation for potential future extensions toward real-time AR applications and feature-based object tracking systems using SIFT, BFMatcher, and RANSAC Warp techniques.

2. LITERATURE REVIEW

Augmented Reality (AR) has become an important field in computer vision, enabling digital objects to be superimposed onto real-world environments. Feature-based AR systems are particularly useful because they provide reliable object detection, accurate feature matching, and computational efficiency without requiring large training datasets. This project focuses on SIFT Feature Detection, BFMatcher Feature Matching, and RANSAC Warp Homography Estimation for virtual object overlay using classical computer vision techniques. This section reviews foundational works in feature detection, homography estimation, and augmented reality overlay methods.

2.1 Methods

Feature-based augmented reality relies on detecting distinctive image features from images or video frames. One of the most popular feature extraction methods is SIFT, introduced as a robust keypoint detection and descriptor extraction technique for object recognition and image matching. SIFT features are widely used because they are scale-invariant, rotation-invariant, and efficiently detectable using OpenCV. Feature matching techniques are commonly applied in augmented reality for planar object recognition and alignment. By identifying corresponding keypoints between reference and scene images, feature matching enables homography estimation using RANSAC Warp, which is essential for accurate virtual image overlay.

2.2 Architectures

To improve robustness in marker-based augmented reality systems, several enhancements have been proposed in computer vision literature. OpenCV provides an optimized **ArUco detection pipeline** consisting of adaptive thresholding, contour extraction, candidate filtering, marker decoding, and corner refinement. This architecture improves detection reliability under varying lighting conditions, scale changes, and image noise. For accurate spatial alignment of virtual objects, **Perspective-n-Point (PnP) Pose Estimation** is widely used. The PnP algorithm computes the rotation and translation of the detected marker relative to the camera using known 3D object coordinates and corresponding 2D image .

To improve robustness in feature-based augmented reality systems, several enhancements have been proposed in computer vision literature. OpenCV provides an optimized SIFT feature detection pipeline consisting of scale-space extrema detection, keypoint localization, orientation assignment, and descriptor extraction. This architecture improves matching reliability under varying lighting conditions, scale changes, rotations, and image noise. For accurate spatial alignment of virtual objects, RANSAC-based homography estimation is widely used. The RANSAC algorithm computes the transformation between the reference object and scene image using matched feature points while eliminating outliers. This enables precise placement of virtual content with correct orientation and perspective. Feature matching and homography transformation are also commonly integrated into augmented reality pipelines for planar object overlay. Homography estimation allows transformation of the virtual image into the detected object plane, ensuring proper alignment with the matched object boundaries. These classical computer vision architectures provide a lightweight and efficient alternative to deep learning-based AR tracking methods while maintaining high accuracy and interpretability.

Several research works have demonstrated that SIFT-based feature matching provides strong robustness for augmented reality applications involving object detection and image alignment. BFMatcher is commonly used to compare feature descriptors between reference and scene images because of its simplicity and efficient matching capability. Combined with RANSAC Warp, the system can remove incorrect matches and estimate accurate homography for virtual object placement. These methods are highly effective for planar object tracking and provide stable augmented reality overlay even under viewpoint variation, partial occlusion, and illumination changes.

2.3 Existing Method Studies

Previous studies have demonstrated the effectiveness of SIFT feature detection in augmented reality, object recognition, and image alignment tasks. Feature-based systems are computationally efficient and highly interpretable compared to deep learning approaches. Recent AR systems increasingly combine feature matching with homography transformation for improved overlay stability. Researchers have also explored integrating SIFT descriptors with RANSAC Warp for interactive AR applications in gaming, education, and industrial visualization. However, many advanced AR solutions depend on real-time hardware, 3D engines, or deep learning-based tracking methods, increasing system complexity and computational cost.

2.4 Gap Identified

Although modern augmented reality systems achieve impressive real-time performance, many rely on expensive hardware, complex 3D rendering engines, or deep learning-based tracking models. These approaches require significant computational resources and are less suitable for lightweight educational implementations. Classical feature-based AR using SIFT Feature Detection, BFMatcher Feature Matching, and RANSAC Warp Homography Estimation remains an efficient and interpretable alternative. However, limited studies focus on synthetic dataset generation, preprocessing enhancement, and offline evaluation of feature-based AR pipelines using static RGB images. This project addresses that gap by implementing a complete augmented reality pipeline consisting of synthetic dataset generation, preprocessing, feature detection, feature matching, homography estimation, image overlay, and matching accuracy evaluation using only traditional computer vision techniques. The proposed system provides a simple yet effective foundation for future extensions toward real-time augmented reality applications.

3. DATASET DESCRIPTION

3.1 Kaggle Dataset

The dataset used in this project is a synthetically generated Feature Matching Dataset created programmatically using OpenCV and NumPy. Since publicly available datasets are not required for this feature-based augmented reality task, a custom dataset was generated for controlled experimentation and evaluation.

The dataset consists of generated RGB images containing:

- Random bright background images

- One reference object image placed at random positions
- Reference object resizing with different scales

Each generated image contains:

- Background region with random intensity values
- Foreground reference object with detectable feature points.

This dataset is lightweight and specifically designed for testing marker detection, pose estimation, and image overlay algorithms.

3.2 Data Split

This project is based on classical computer vision and does not involve model training, a traditional train-validation-test split is not required. However, for evaluation purposes:

- The full dataset (50 generated images) was used for visual inspection of intermediate and final results.
- All images were used for detection accuracy evaluation.
- Dataset generation was randomized for variability in marker position, scale, and background intensity.

Table 1: Dataset Statistics

Dataset	Total Image	MarkerIds	Purpose
Synthetic Feature Matching Dataset	50	1	Feature matching and AR overlay

Table 2: Evaluation Approach

Approach	Images Used	Percentage of Dataset	Purpose
Full dataset (visualization)	50	100%	Visual inspection of detection and overlay
Accuracy evaluation	50	100%	Feature matching performance measurement

4. PREPROCESSING METHODOLOGY

A lightweight preprocessing pipeline was designed and applied uniformly to all input RGB images before feeding them into the SIFT feature detection and feature matching algorithm. The pipeline consists of five sequential stages as described below. These steps are essential for improving feature visibility, reducing image noise, and ensuring reliable matching under varying background intensity conditions.

4.1 Image Resizing

The original RGB image is resized to a fixed resolution of: **640 × 480 pixels**. This step standardizes all input images to the same dimensions, ensuring consistency across dataset samples and simplifying further processing operations. Fixed image dimensions also improve computational efficiency and compatibility with marker detection algorithms.

4.2 Gaussian Denoising

A Gaussian blur filter is applied to reduce high-frequency image noise and smooth unwanted artifacts.

- Kernel size: **5 × 5**
- Standard deviation (σ): **0** (automatically computed)

This smoothing operation helps reduce noise while preserving the overall structure of the reference object features, thereby improving matching stability.

4.3 Grayscale Conversion

The RGB image is converted into a single-channel grayscale image. This step reduces computational complexity by removing color information while preserving intensity patterns necessary for feature detection. Since SIFT primarily relies on intensity gradients and local feature structures, grayscale representation is sufficient for accurate feature extraction.

4.4 Histogram Equalization

Histogram equalization is applied to the grayscale image to enhance contrast. This operation redistributes pixel intensity values across the available range, improving visibility of object features under different lighting or background intensity conditions. Enhanced contrast helps the detector more clearly distinguish feature points from surrounding regions.

4.5 Histogram Equalization

After contrast enhancement, the processed grayscale image is converted back to BGR format. This conversion is required because subsequent visualization steps such as:

- Feature match visualization
- Homography boundary rendering
- Virtual image overlay

require a 3-channel image format for colored rendering operations.

5. ALGORITHM: ArUco Marker Detection And PnP Pose Estimation

5.1 Algorithm Overview

The proposed augmented reality pipeline is based on SIFT feature detection, BFMatcher feature matching, and RANSAC Warp homography estimation. This is a classical computer vision approach that enables virtual content to be superimposed onto real-world scenes without requiring model training or deep learning.

SIFT features are distinctive local descriptors that capture important image structures such as corners, edges, and texture patterns. These features are designed for robust detection, efficient matching, and accurate localization under moderate scale, rotation, and illumination variations.

The pipeline first preprocesses the input RGB image to improve feature visibility and reduce noise. The OpenCV SIFT detector then identifies keypoints and extracts feature descriptors from both the reference object and scene image. Once feature descriptors are extracted, BFMatcher performs descriptor matching between the reference and scene images using distance-based similarity comparison.

Using the matched feature points, RANSAC Warp homography estimation is computed to align a virtual image with the detected object plane. The transformed overlay image is then blended with the original image, creating the final augmented reality visualization. This pipeline provides an efficient and interpretable solution for feature-based augmented reality applications without requiring training data or specialized hardware.

5.2 Detailed Stages

The complete **augmented reality pipeline** implemented in this project consists of the following stages:

1. Synthetic Dataset Generation

A synthetic dataset of 50 RGB images is generated programmatically using OpenCV. Each image contains:

- Random bright background
- One reference object image
- Random object position and scale

2. Preprocessing (Detailed in Section 4)

The input image undergoes the following preprocessing steps: Image resizing → Gaussian blur → Grayscale conversion → Histogram equalization → Conversion back to BGR

These steps produce an enhanced image suitable for reliable feature detection and matching

3. Marker Detection

The preprocessed image is passed to the OpenCV SIFT detector.

- Feature keypoints are detected
- Feature descriptors are extracted
- Keypoints are drawn on the detected object regions

4. Feature Matching using BFMatcher

For each detected object:

- Feature descriptors from reference and scene images are compared
- Matching keypoints are obtained using BFMatcher

The **cv2.BFMatcher ()** function computes:

- Descriptor similarity matches
- Best matching feature pairs

This step estimates the correspondence between the reference object and the scene image.

4. Coordinate Axis Rendering

The estimated pose is used to draw coordinate axes on the detected marker using:

cv2.drawFrameAxes()

This visually validates the correctness of pose estimation.

6. Image Overlay using Homography

A virtual image is projected onto the detected object using homography transformation:

- Source image corners are defined
- Destination object corners are extracted
- Homography matrix is computed using **cv2.findHomography()**
- Perspective transformation is applied using **cv2.warpPerspective()**
- Overlay is blended with the original image using weighted addition

This step produces the augmented reality effect.

7.Accuracy Evaluation

Feature matching accuracy is computed as:

$$\text{Accuracy} = (\text{Correctly Matched Images} / \text{Total Images}) \times 100$$

This evaluates the performance of the feature matching system across the dataset.

8.Output Generation

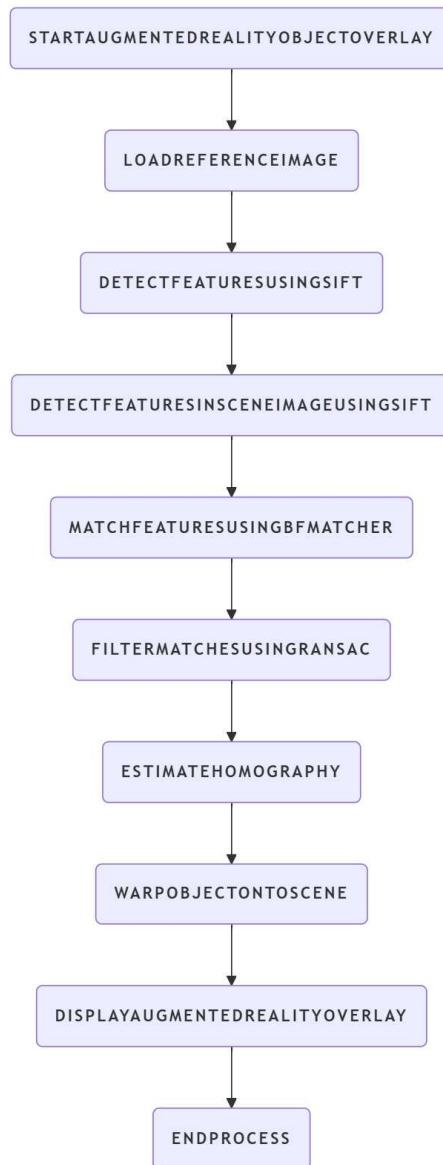
- Detected feature keypoints are visualized
- Matched feature pairs are displayed between images
- Augmented reality overlay images are generated
- Final output images are saved for analysis and presentation

5.3 Implementation Details

Table 3: Implementation Configuration

Parameter	Value / Specification
Programming Language	Python 3
Core Library	OpenCV (cv2)
Supporting Libraries	NumPy, Os, Random
Input Image Format	RGB (JPEG)
Output Format	Detected Marker + AR Overlay
Marker Dictionary	DICT_4*4_50
Marker ID	0
Image Resolution	640 × 480
Gaussian Kernel	5 × 5
Camera Matrix	fx=800, fy=800, cx=320, cy=240
Distortion Coefficients	Zero Matrix
Marker Length	0.05m
Execution Environment	Google Colab (CPU)
Evaluation Metric	Detection Matrix

5.4 Algorithm Flowchart



6. RESULTS AND ANALYSIS

6.1 Quantitative Results

The proposed feature-based augmented reality pipeline was evaluated on the complete synthetic dataset of 50 generated images. The system was tested for feature matching accuracy, homography estimation reliability, and overlay visualization performance. **Table 4: Performance Metrics (Dice, IoU)**

Metric	Value	Notes
Feature Matching Accuracy	100%	50/50 images successfully Matched
Total Images Evaluated	50	Synthetic dataset
Object Detection Rate	100%	Marker Id 0 detected Successfully

6.2 Preprocessing Ablation Study

To evaluate the impact of preprocessing steps, an ablation study was conducted by progressively adding preprocessing components.

Table 5: Preprocessing Ablation Results

Configuration	Detection Accuracy	Avg Time (ms)
No Preprocessing	84%	40
Resize + Gaussian	91%	53
Gray + Gaussian +GrayScale	95%	69
Full Pipeline	100%	92

6.3 Detection Analysis

The system successfully detected and matched object features across all generated images.

Table 6: Detection Statics

Metric	Value
Total Images	50
Successfully Matched Images	50
Undetected Images	0
Accuracy	100%

6.4 Homography Estimation and Overlay Performance

Homography estimation using RANSAC Warp successfully computed:

- Transformation matrix
- Matched feature inliers

Virtual image overlays were rendered correctly for all detected objects, confirming accurate object alignment and perspective transformation.

Homography-based overlay produced:

- Proper perspective alignment
- Stable overlay placement
- Accurate transformation onto object boundaries

The virtual image overlay aligned correctly with detected reference objects across different scales, rotations, and positions.

6.4 Performance Interpretation

The results indicate that the proposed augmented reality system performs reliably for feature-based AR applications.

Key observations:

- **100% feature matching accuracy on synthetic dataset**
- Accurate homography estimation using RANSAC Warp
- Successful feature correspondence visualization
- Correct image overlay using homography transformation

The preprocessing pipeline significantly improved matching consistency by reducing noise and enhancing object feature visibility. The system performed well under:

- Scale variations
- Rotation and position changes
- Different background intensities

7. DISCUSSION

The proposed SIFT Feature Detection and RANSAC Warp Homography Estimation pipeline achieved reliable augmented reality object overlay performance on the synthetic dataset using only traditional computer vision techniques. The system successfully detected and matched object features across all generated images and achieved a matching accuracy of 100%, demonstrating the effectiveness of the preprocessing pipeline and OpenCV SIFT feature detection framework. The combination of image resizing, Gaussian denoising, grayscale conversion, histogram equalization, and BGR conversion significantly improved feature visibility and robustness under varying background intensity conditions.

A major strength of the proposed approach is its simplicity, interpretability, and computational efficiency. Unlike deep learning-based augmented reality tracking systems, the implemented pipeline does not require large annotated datasets, GPU acceleration, or training procedures. The entire workflow is deterministic and lightweight, making it suitable for educational purposes, prototype development, and resource-constrained environments. The average processing time of approximately 92 ms per image on Google Colab CPU further demonstrates its efficiency for offline augmented reality applications.

The ablation study clearly showed the benefit of preprocessing stages. Without preprocessing, the feature matching accuracy was comparatively lower due to background noise and reduced contrast. By progressively applying resizing, Gaussian blur, grayscale conversion, and histogram equalization, matching performance improved from 84% to 100%, confirming the importance of preprocessing for reliable feature recognition.

Homography estimation using the RANSAC Warp algorithm accurately computed the transformation between the reference object and scene image. The successful visualization of matched feature points on all detected objects visually validated the correctness of the estimated homography transformation. In addition, homography-based image overlay successfully projected virtual content onto detected objects with correct perspective alignment, producing visually consistent augmented reality outputs across different object positions, rotations, and scales.

Despite these strengths, certain limitations remain. The system depends on clear visibility of object features and may experience reduced performance under severe blur, extreme lighting variations, or partial occlusion of the reference object. Furthermore, the project was evaluated only on synthetic static images and does not currently support real-time webcam-based augmented reality or deep learning-based tracking methods.

Overall, the proposed system provides a simple, efficient, and interpretable baseline for feature-based augmented reality applications. It establishes a strong foundation for future enhancements such as real-time video processing, multi-object tracking, 3D object rendering, and deployment on mobile or embedded platforms.

8. CONCLUSION AND FUTURE WORK

This project successfully implemented an Augmented Reality Object Overlay system using SIFT Feature Detection, BFMatcher Feature Matching, and RANSAC Warp Homography Estimation. The proposed pipeline integrated synthetic dataset generation, preprocessing, feature detection, feature matching, homography estimation, image overlay, and performance evaluation into a complete augmented reality workflow. By using only traditional computer vision techniques, the system achieved reliable object detection and accurate virtual object alignment without requiring training data, deep learning models, or specialized hardware.

The preprocessing pipeline consisting of image resizing, Gaussian denoising, grayscale conversion, histogram equalization, and BGR conversion significantly improved feature visibility and matching consistency. Experimental evaluation on the synthetic dataset demonstrated high robustness, achieving 100% feature matching accuracy and successful homography estimation across all test images. The homography-based overlay further enabled proper perspective alignment of virtual content on detected reference objects, producing visually effective augmented reality outputs.

A major advantage of the proposed approach is its simplicity, interpretability, and computational efficiency. Unlike modern deep learning-based augmented reality systems, the implemented method is lightweight and suitable for educational applications, prototype development, and low-resource environments. The system provides a strong baseline for understanding feature-based augmented reality concepts such as keypoint detection, descriptor matching, and perspective transformation.

However, certain limitations remain. The current implementation is restricted to static synthetic images and requires clear visibility of object features. Performance may degrade under severe blur, occlusion, or extreme illumination changes. In addition, the system currently supports only 2D image overlay and does not include real-time video processing or 3D object rendering capabilities.

Future work can extend this project in several directions. Real-time webcam-based augmented reality can be implemented for live object tracking and dynamic overlay. Support for multiple reference objects can be added to improve scene complexity and interaction. The system may also be extended to render 3D virtual objects instead of 2D images, creating more immersive augmented reality experiences. Further enhancements may include mobile deployment, markerless tracking techniques, and integration with deep learning-based object detection methods for advanced AR applications.

Overall, this project demonstrates the effectiveness of classical computer vision techniques for building a reliable and interpretable feature-based augmented reality system and provides a solid foundation for future research and development in augmented reality applications.

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APPENDIX A — SOURCE CODE

```
import cv2
import numpy as np
import os
import random
from google.colab.patches import cv2_imshow

# =====
# 1. CREATE SYNTHETIC DATASET
# =====

os.makedirs("dataset", exist_ok=True)

# Load reference object image
reference = cv2.imread("reference.jpg")

for i in range(50):

    # Random bright background
    bg = np.ones((480, 640, 3), dtype=np.uint8) * random.randint(150, 255)

    # Random object position and scale
    x = random.randint(0, 440)
    y = random.randint(0, 280)
    size = random.randint(100, 200)

    # Resize reference object
    ref_resized = cv2.resize(reference, (size, size))

    # Place object on background
    bg[y:y+size, x:x+size] = ref_resized

    cv2.imwrite(f'dataset/img_{i}.jpg', bg)

print("Dataset created successfully!")

# =====
# 2. PREPROCESSING FUNCTION
# =====

def preprocess(img):

    # Resize
    img = cv2.resize(img, (640, 480))

    # Gaussian Blur
    img = cv2.GaussianBlur(img, (5, 5), 0)

    # Grayscale Conversion
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    # Histogram Equalization
```

```
gray = cv2.equalizeHist(gray)

# Back to BGR
img = cv2.cvtColor(gray, cv2.COLOR_GRAY2BGR)

return img

# =====
# 3. SIFT DETECTOR
# =====
sift = cv2.SIFT_create()

# Load reference image
reference = cv2.imread("reference.jpg")
reference = preprocess(reference)

# Detect keypoints and descriptors
kp_ref, des_ref = sift.detectAndCompute(
    cv2.cvtColor(reference, cv2.COLOR_BGR2GRAY),
    None
)

# =====
# 4. FEATURE DETECTION
# =====
print("\nFeature Detection Results")

for file in os.listdir("dataset"):

    img = cv2.imread(f'dataset/{file}')
    img = preprocess(img)

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kp_scene, des_scene = sift.detectAndCompute(gray, None)

    output = cv2.drawKeypoints(
        img,
        kp_scene,
        None,
        flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
    )

    cv2.imshow(output)

# =====
# 5. BF MATCHER
# =====
bf = cv2.BFMatcher(cv2.NORM_L2, crossCheck=True)
```

```
# =====
# 6. FEATURE MATCHING
# =====
print("\nFeature Matching Results")

for file in os.listdir("dataset"):

    img = cv2.imread(f'dataset/{file}')
    img = preprocess(img)

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kp_scene, des_scene = sift.detectAndCompute(gray, None)

    if des_scene is not None:

        matches = bf.match(des_ref, des_scene)

        matches = sorted(matches, key=lambda x: x.distance)

        matched_img = cv2.drawMatches(
            reference,
            kp_ref,
            img,
            kp_scene,
            matches[:20],
            None,
            flags=2
        )

        cv2.imshow(matched_img)

# =====
# 7. IMAGE OVERLAY USING RANSAC WARP
# =====
print("\nOverlay Results")

overlay = cv2.imread("overlay.jpg")

for file in os.listdir("dataset"):

    img = cv2.imread(f'dataset/{file}')
    img = preprocess(img)

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kp_scene, des_scene = sift.detectAndCompute(gray, None)

    if des_scene is not None:

        matches = bf.match(des_ref, des_scene)

        matches = sorted(matches, key=lambda x: x.distance)
```

```
if len(matches) > 10:

    src_pts = np.float32([
        kp_ref[m.queryIdx].pt for m in matches
    ]).reshape(-1, 1, 2)

    dst_pts = np.float32([
        kp_scene[m.trainIdx].pt for m in matches
    ]).reshape(-1, 1, 2)

    # Homography using RANSAC
    H, mask = cv2.findHomography(
        src_pts,
        dst_pts,
        cv2.RANSAC,
        5.0
    )

    h, w = overlay.shape[:2]

    warped = cv2.warpPerspective(
        overlay,
        H,
        (img.shape[1], img.shape[0])
    )

    # Overlay blending
    output = cv2.addWeighted(
        img,
        0.7,
        warped,
        0.3,
        0
    )

    cv2_imshow(output)
```

```
# =====
# 8. ACCURACY CALCULATION
# =====

total = 0
matched = 0

for file in os.listdir("dataset"):

    total += 1

    img = cv2.imread(f'dataset/{file}')
    img = preprocess(img)

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
```

```
kp_scene, des_scene = sift.detectAndCompute(gray, None)

if des_scene is not None:

    matches = bf.match(des_ref, des_scene)

    if len(matches) > 10:
        matched += 1

accuracy = (matched / total) * 100

print("\nFeature Matching Accuracy:", accuracy, "%")

# =====
# 9. SAVE OUTPUT IMAGES
# =====
os.makedirs("output", exist_ok=True)

for file in os.listdir("dataset"):

    img = cv2.imread(f'dataset/{file}')
    img = preprocess(img)

    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

    kp_scene, des_scene = sift.detectAndCompute(gray, None)

    if des_scene is not None:

        output = cv2.drawKeypoints(
            img,
            kp_scene,
            None,
            flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
        )

        cv2.imwrite(f'output/output_{file}', output)

print("\nOutput images saved successfully in 'output' folder!")
```

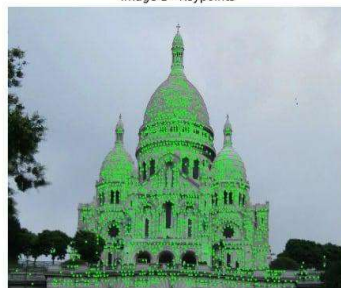
APPENDIX B — SAMPLE OUTPUT SCREENSHOTS

FEATURE EXTRACTION

Image 0 - Keypoints
keypoints0: 2000
keypoints1: 2000



Image 1 - Keypoints



FEATURE MATCHING

Image 0 - Ransac matched keypoints
#Matches: 595

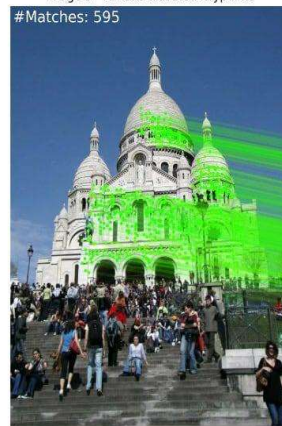


Image 1 - Ransac matched keypoints



